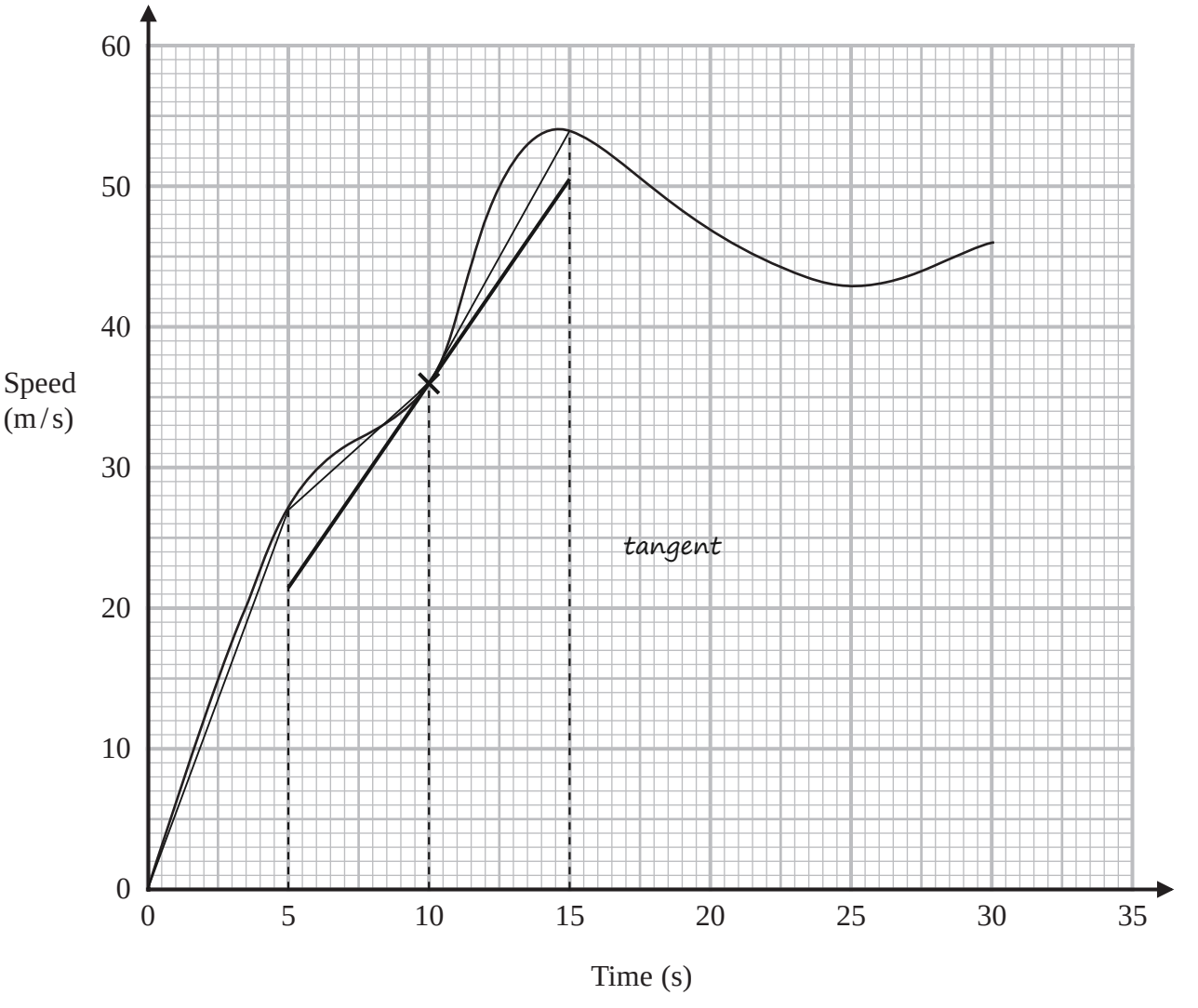
15 Here is the speed-time graph of a racing car for the first 30 seconds of a race.



(a) Work out an estimate for the acceleration, in m/s^2, of the racing car 10 seconds after the start of the race.
You must show how you get your answer.

Tangent points ≈ (5, 21.5) and (15, 50.6).

Acceleration = (50.6 – 21.5) ÷ (15 – 5)

= 29.1 ÷ 10 = 2.91 m/s^2 ≈ 2.9 m/s^2

2.9 ..... m/s^2

(3)

1 mark
Draw a tangent touching the ACTUAL curve at t = 10.

2 marks total
Gradient = change in speed ÷ change in time on that tangent.

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(b) Work out an estimate for the distance travelled by the racing car in the first 15 seconds of the race.
Use 3 strips of equal width.

Read speeds at 0, 5, 10, 15 s: 0, 27, 36, 54 m/s.

$\frac{1}{2} \times (0 + 27) \times 5 = 67.5$ m

$\frac{1}{2} \times (27 + 36) \times 5 = 157.5$ m

$\frac{1}{2} \times (36 + 54) \times 5 = 225$ m

Distance $\approx 67.5 + 157.5 + 225 = 450$ m

450 m

(3)

(Total for Question 15 is 6 marks)

1 mark
Each trapezium: half × sum of parallel sides × width 5 s.

2 marks total
Add all THREE trapezia; area under speed–time graph is distance.

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